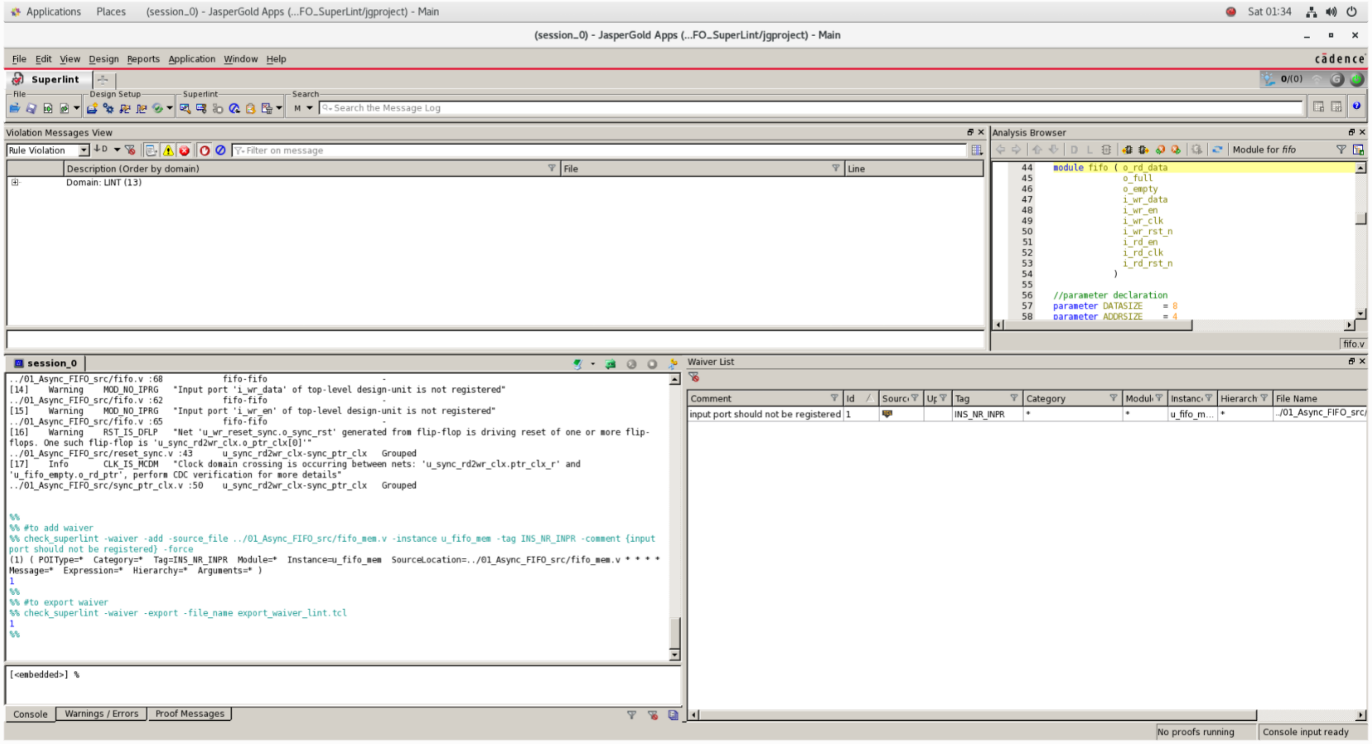
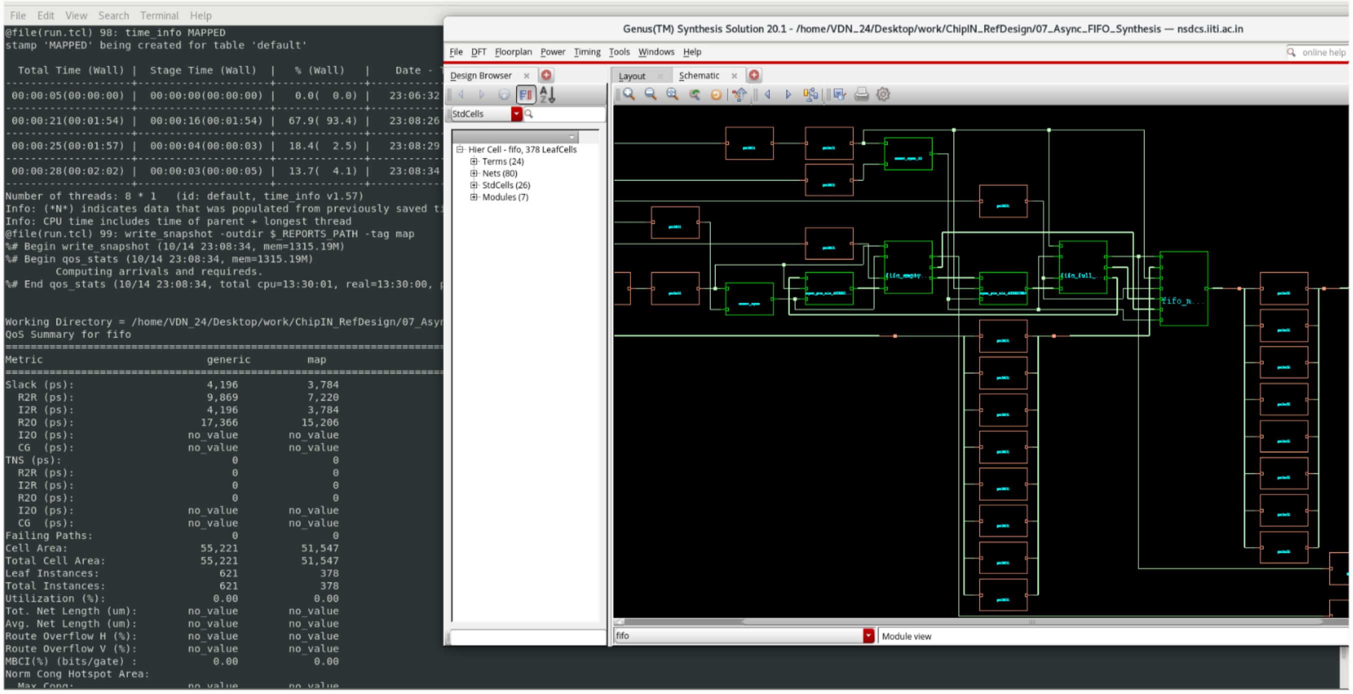
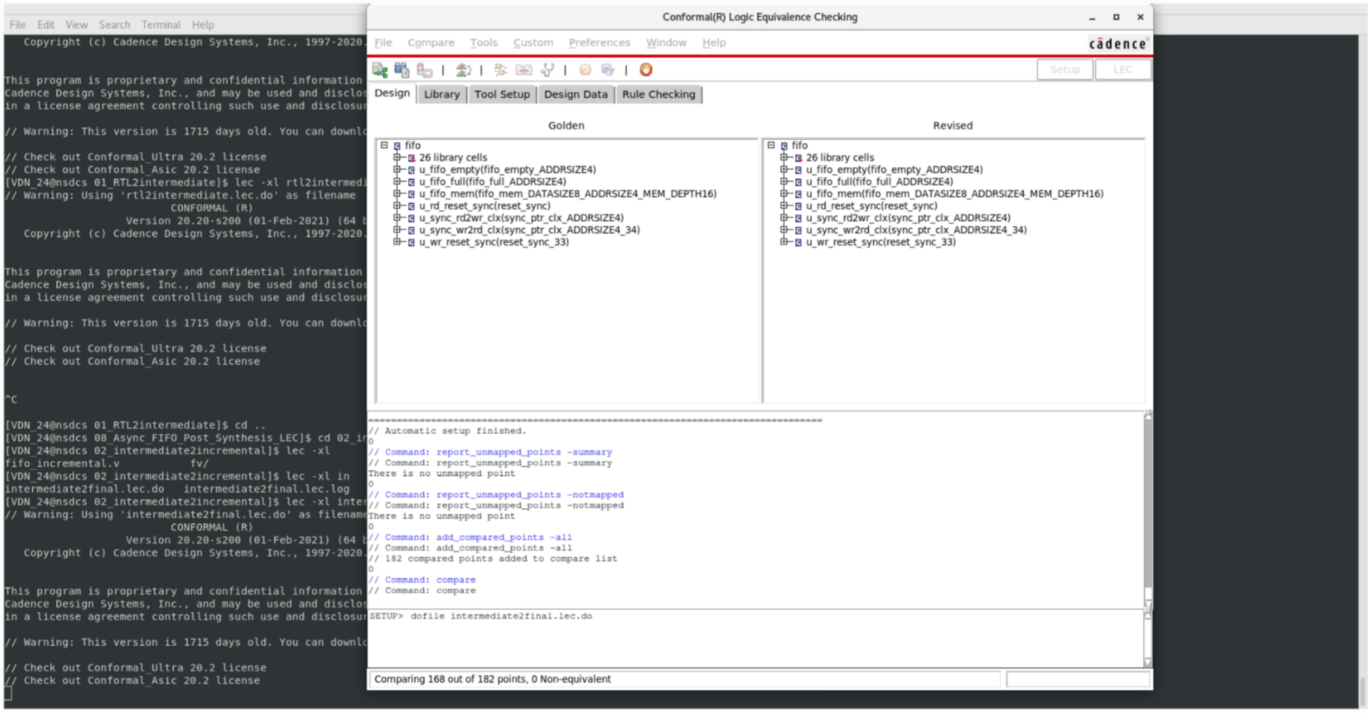
# FIFO Lint



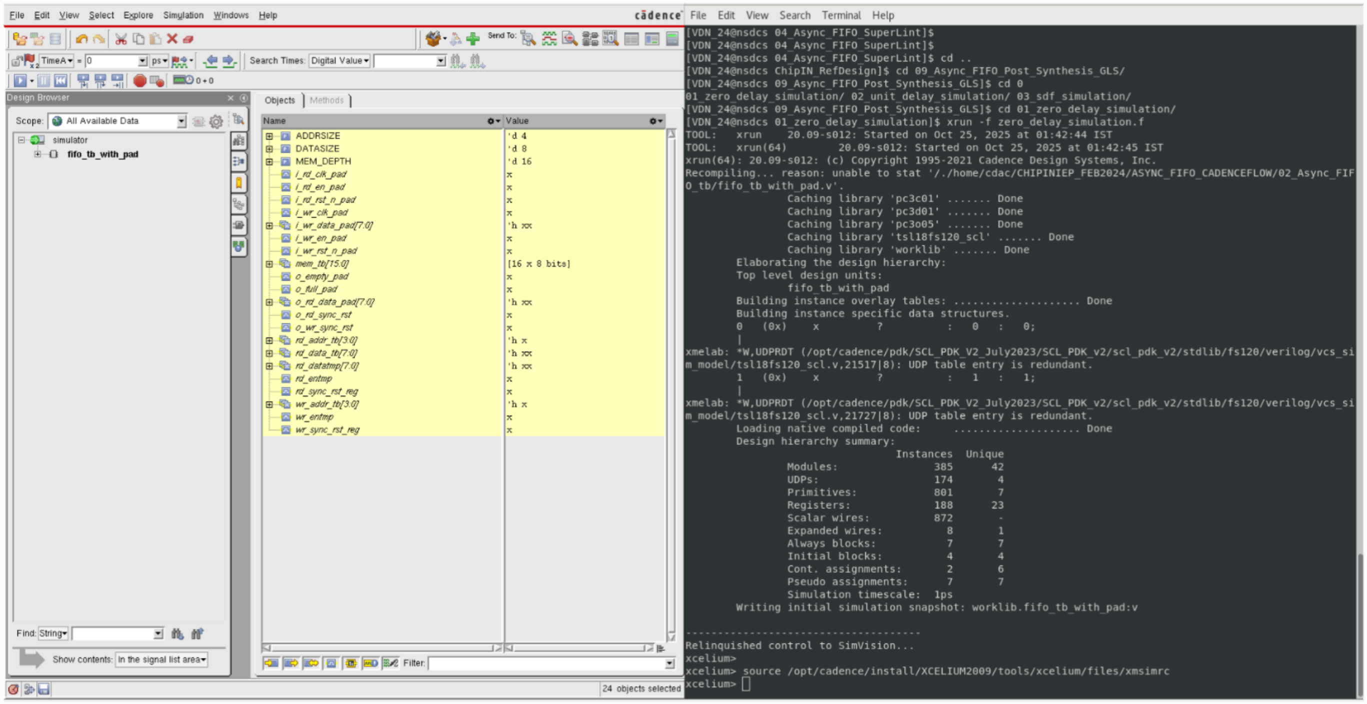
# FIFO Synthesis

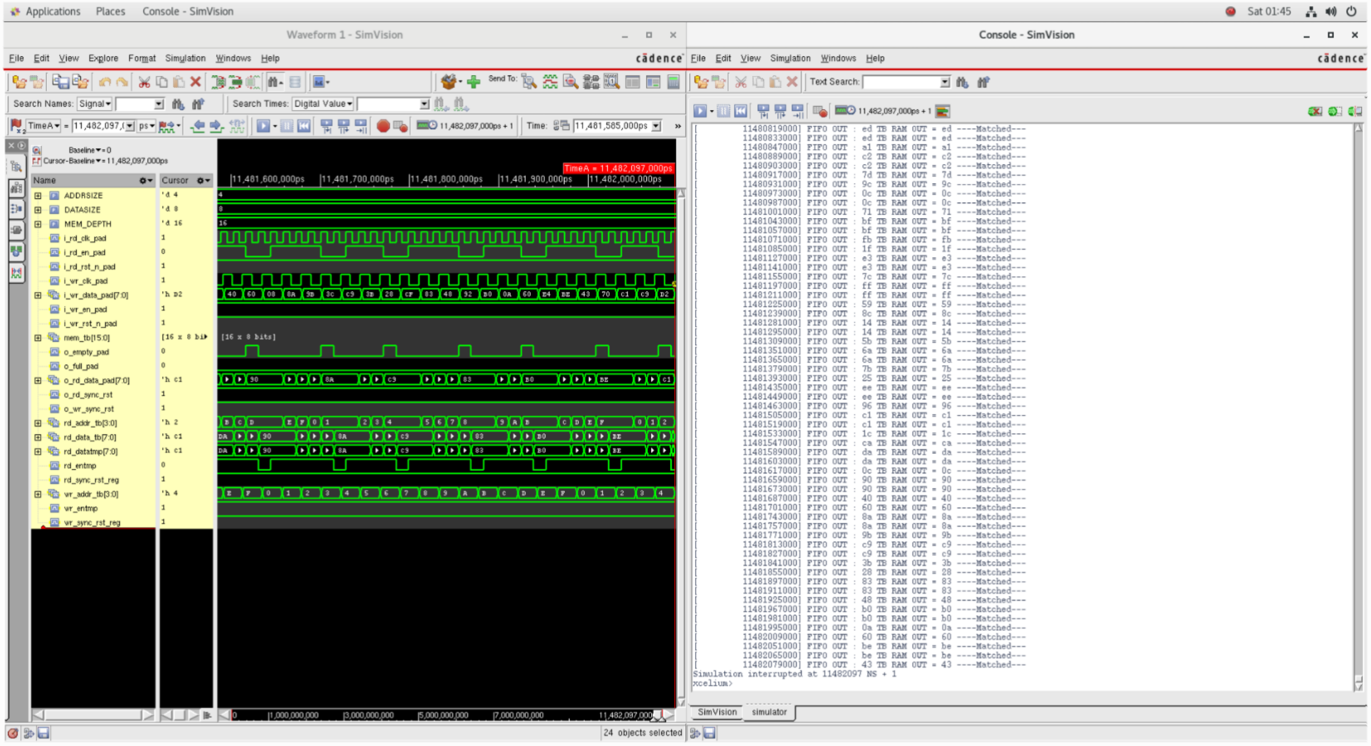


# FIFO Post Synthesis

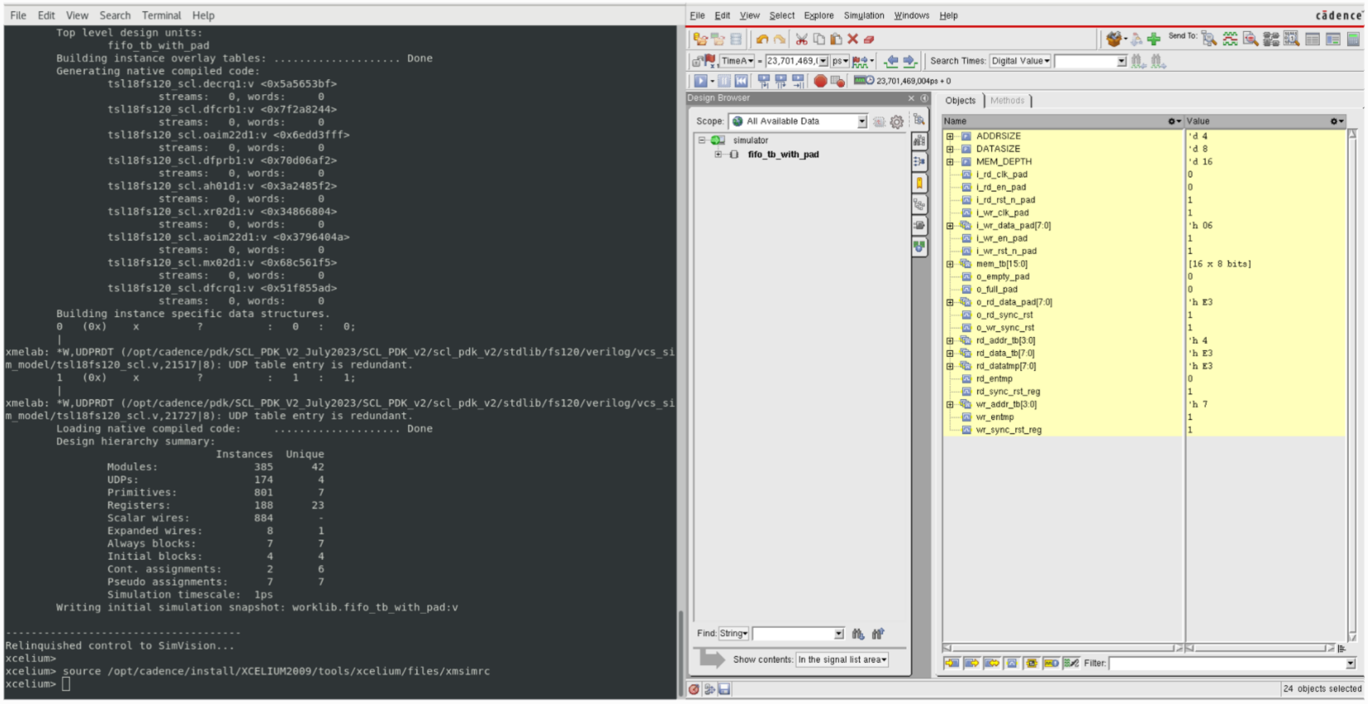


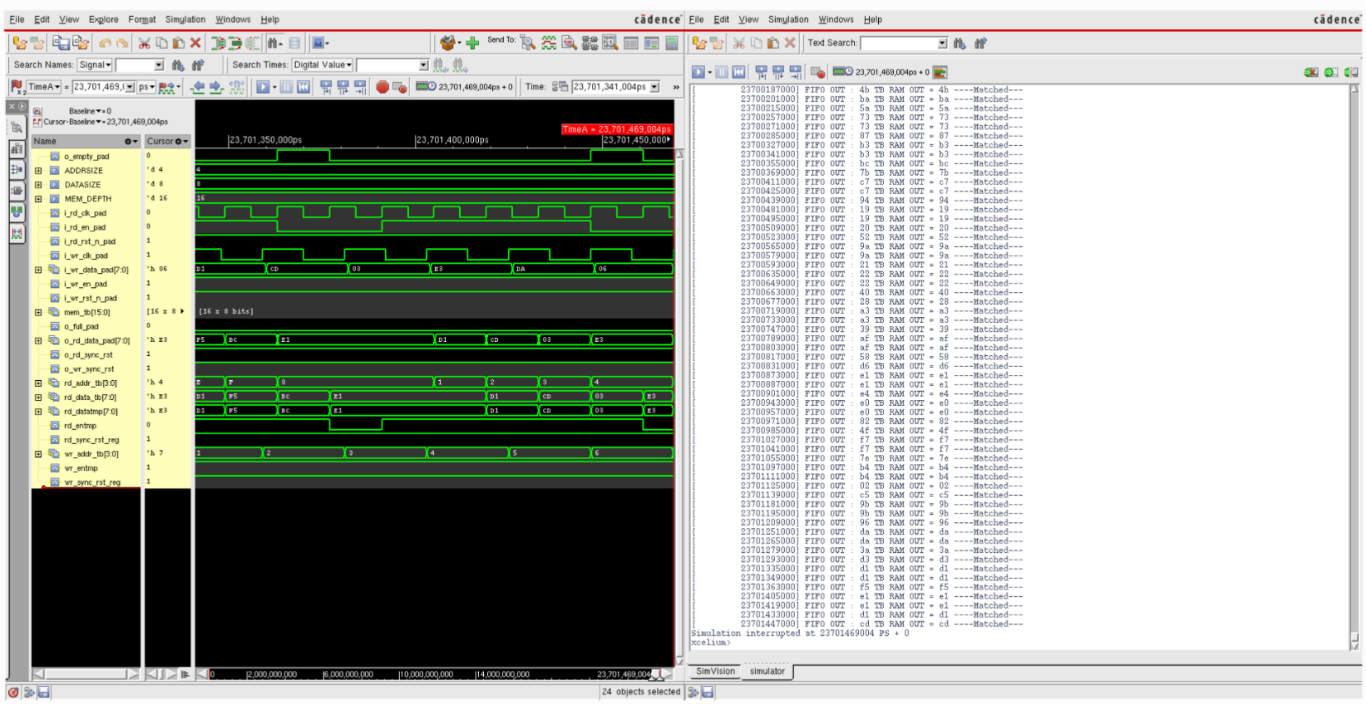
# Zero_delay_simulation



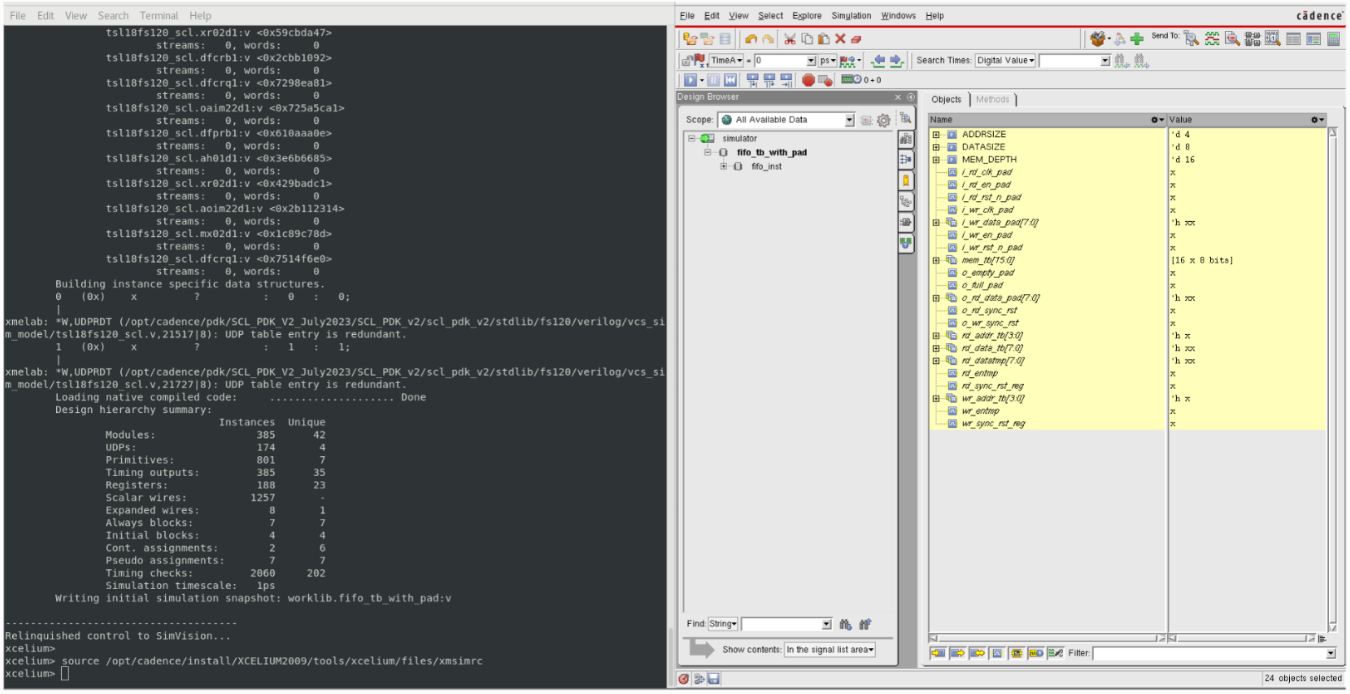



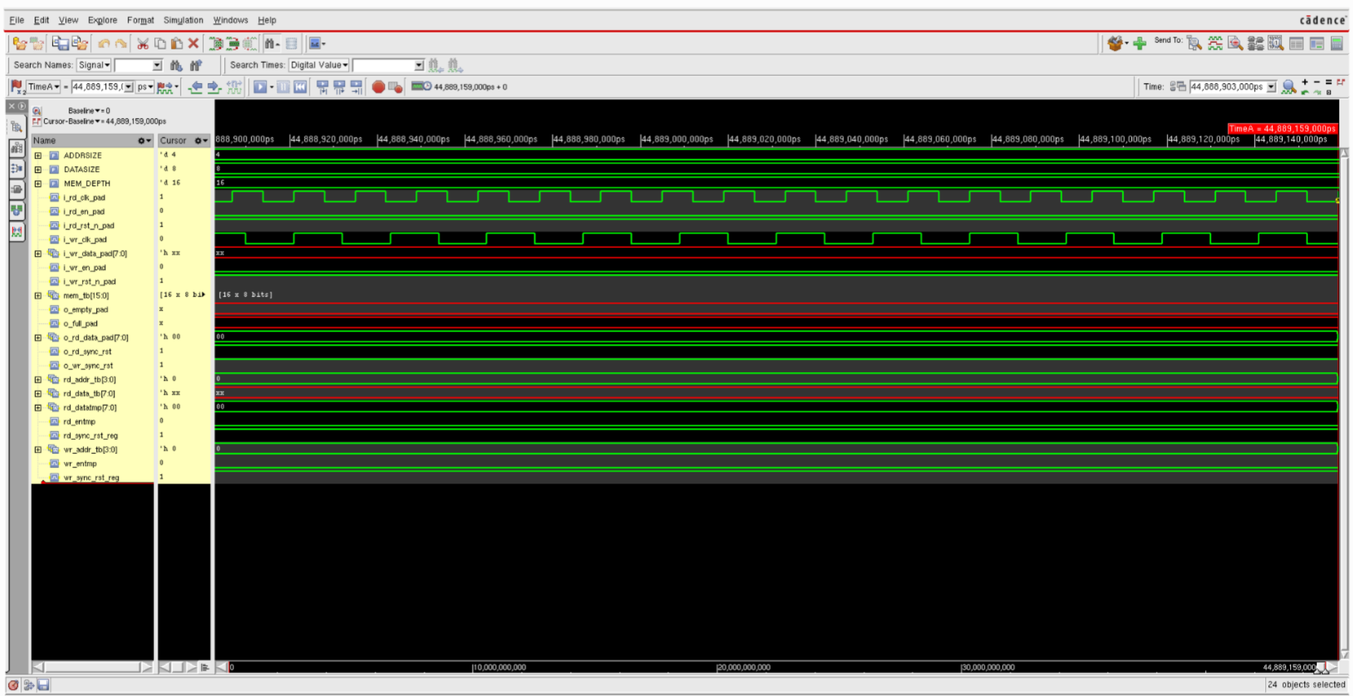
# Unit_delay Simulation



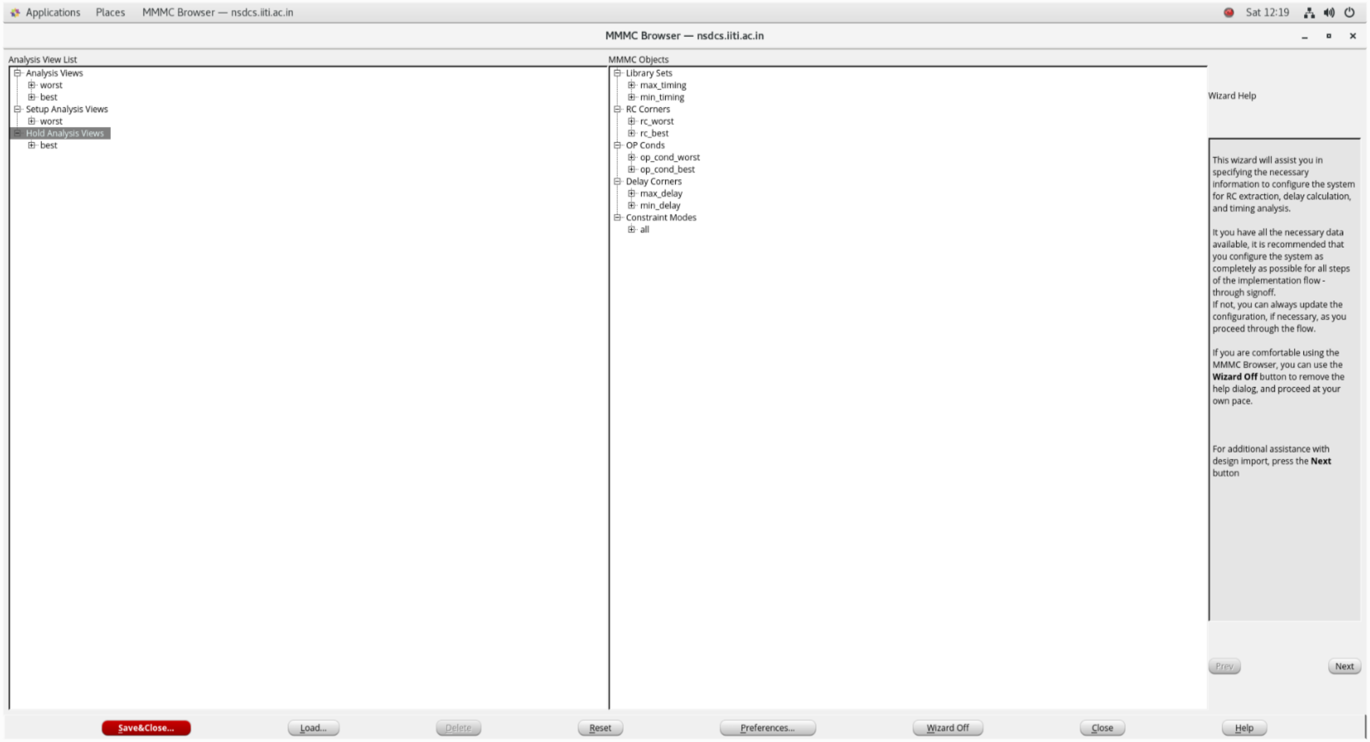



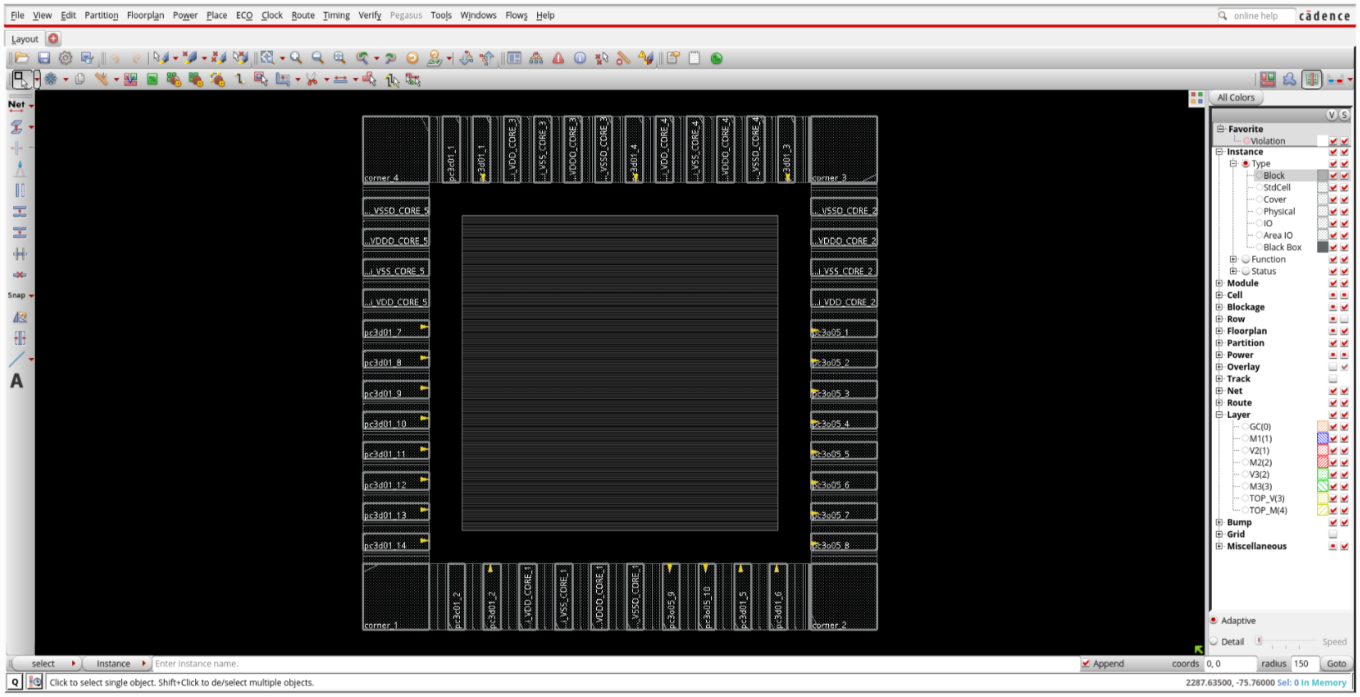
# Sdf_ simulation



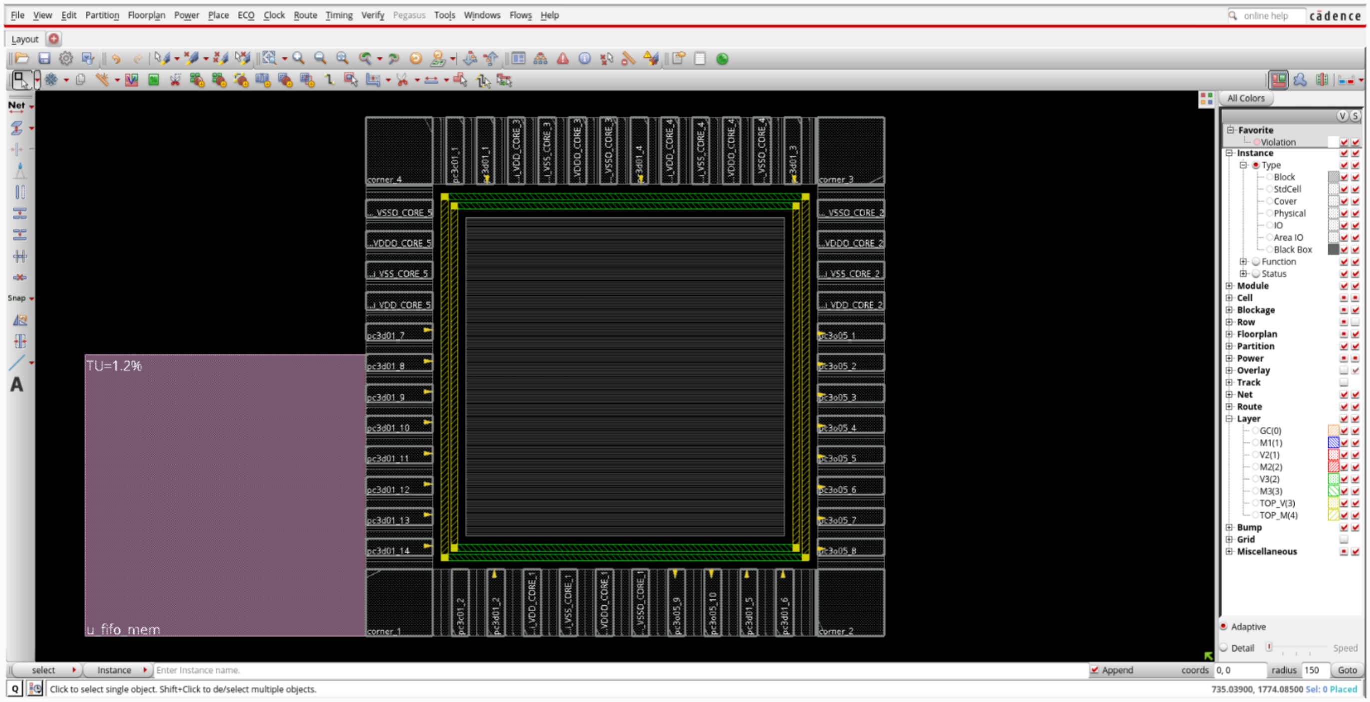


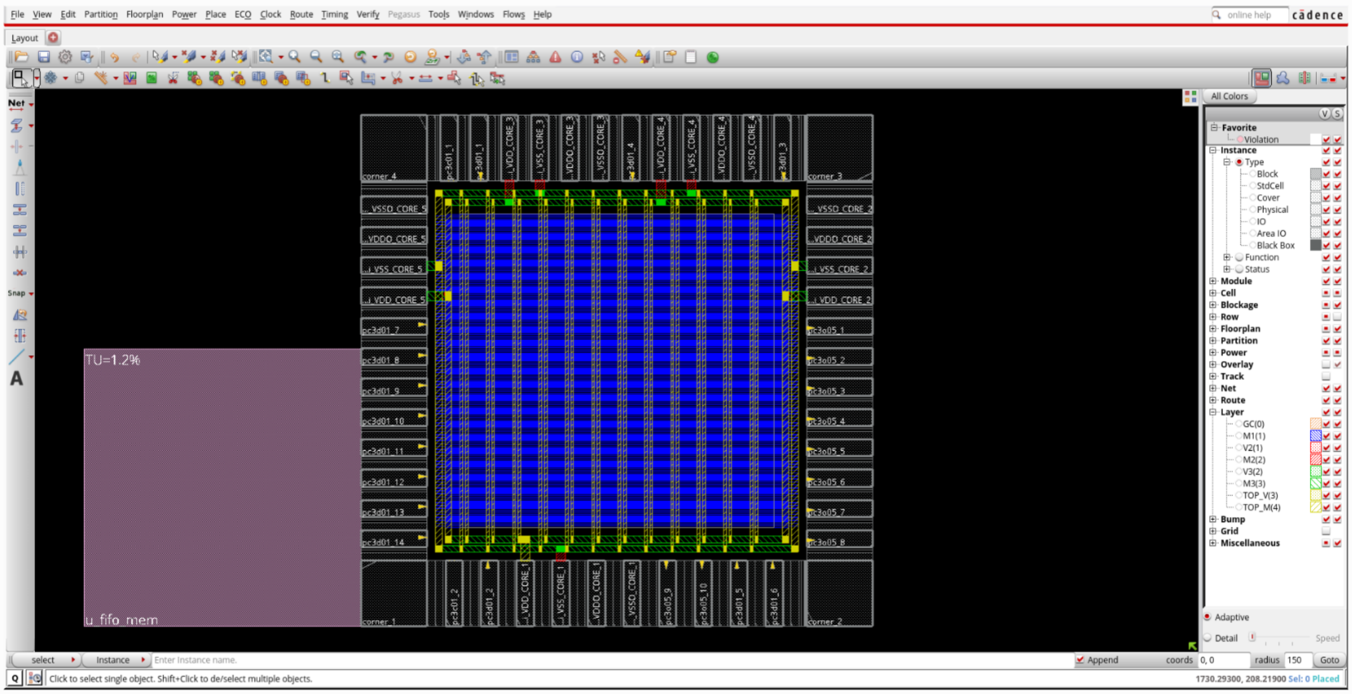
# FIFO Floorplanning



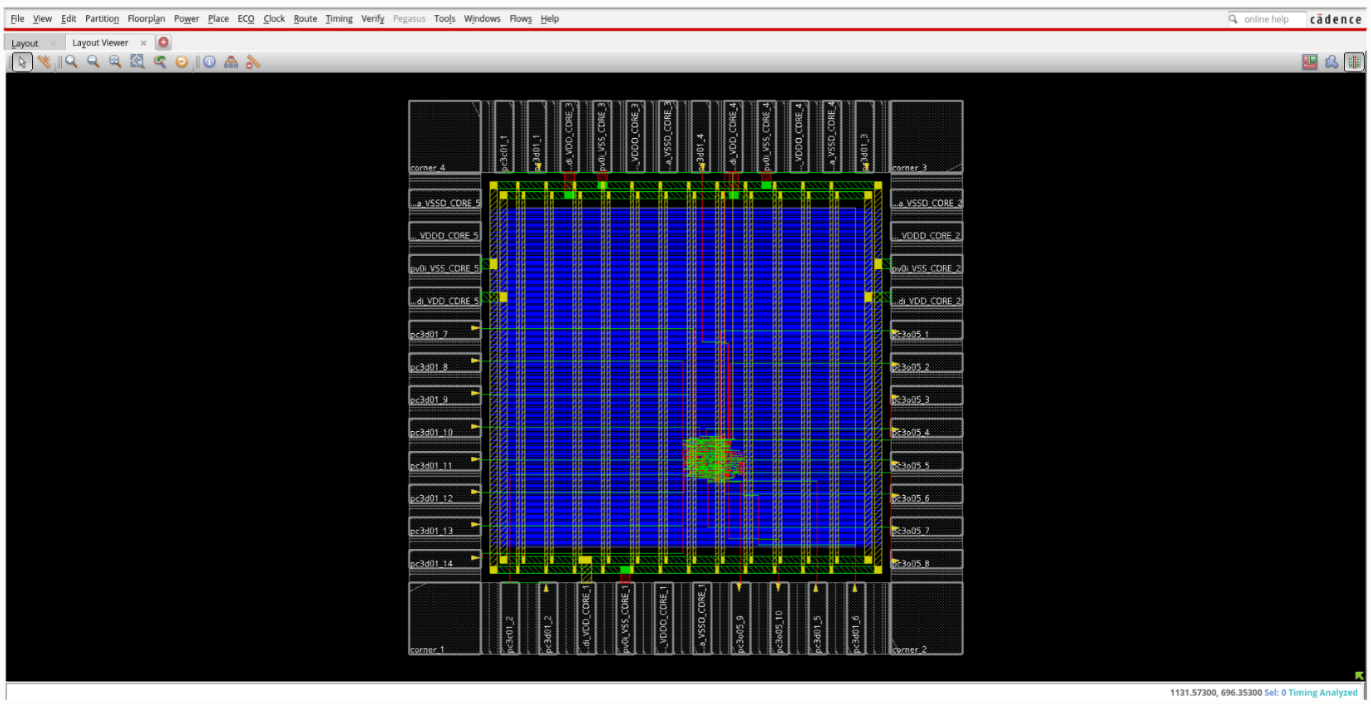
# FIFO Power Planning



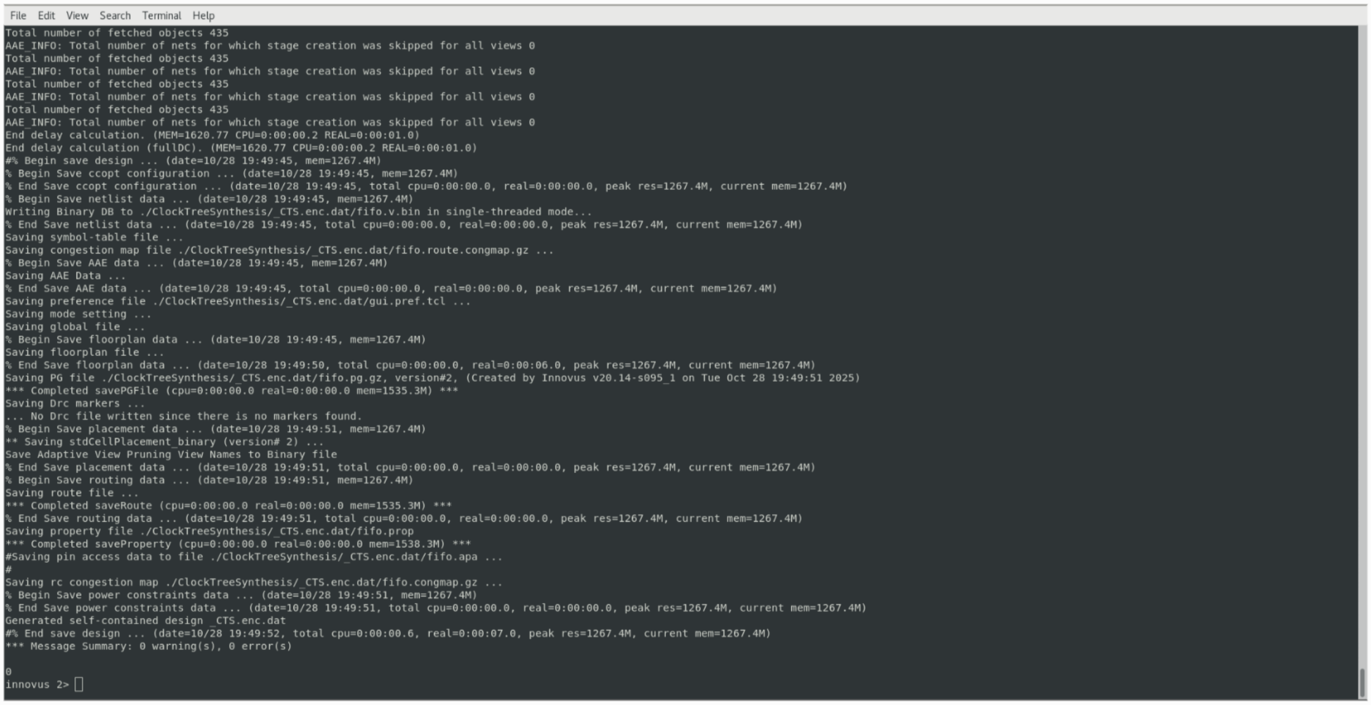


# FIFO Placement

```
File  Edit  View  Search  Terminal  Help
**place_opt_design ... cpu = 0:00:19, real = 0:00:26, mem = 1466.2M **
*** Finished GigaPlace ***

*** Summary of all messages that are not suppressed in this session:
Severity  ID               Count  Summary
WARNING   IMPEXT-6166          3  Capacitance table file(s) without the EX...
WARNING   IMPEXT-3530          3  The process node is not set. Use the com...
WARNING   IMPSP-9025           1  No scan chain specified/traced.
*** Message Summary: 7 warning(s), 0 error(s)

*** place_opt_design #1 [finish] : cpu/real = 0:00:19.2/0:00:25.9 (0.7), totSession cpu/real = 0:01:50.2/0:05:55.0 (0.3), mem = 1466.2M
innovus 1> *** timeDesign #2 [begin] : totSession cpu/real = 0:01:53.0/0:06:06.8 (0.3), mem = 1476.4M
Start to check current routing status for nets...
All nets are already routed correctly.
End to check current routing status for nets (mem=1466.4M)

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Setup views included:
 worst

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  5.203  | 10.016  |  5.203  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   465   |   328   |   137   |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                | Nr nets(terms)   | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 1.255%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------------
Reported timing to dir timingReports
Total CPU time: 0.52 sec
Total Real time: 12.0 sec
Total Memory Usage: 1475.167969 Mbytes
*** timeDesign #2 [finish] : cpu/real = 0:00:00.5/0:00:11.8 (0.0), totSession cpu/real = 0:01:53.5/0:06:18.6 (0.3), mem = 1475.2M
innovus 1>
```

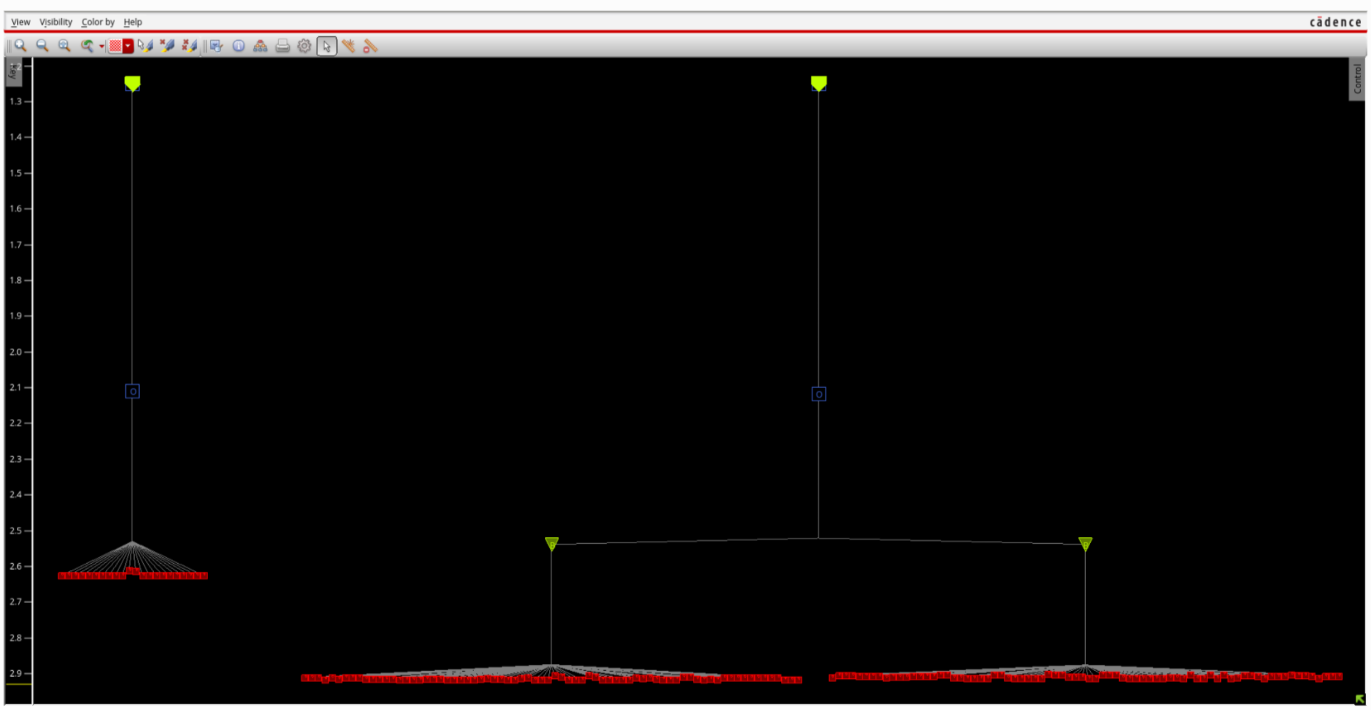


# FIFO Clock Tree synthesis



File Edit View Search Terminal Help

```
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=1620.77 CPU=0:00:00.2 REAL=0:00:01.0)
End delay calculation (fullDC). (MEM=1620.77 CPU=0:00:00.2 REAL=0:00:01.0)
#% Begin save design ... (date=10/28 19:49:45, mem=1267.4M)
% Begin Save ccopt configuration ... (date=10/28 19:49:45, mem=1267.4M)
% End Save ccopt configuration ... (date=10/28 19:49:45, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
% Begin Save netlist data ... (date=10/28 19:49:45, mem=1267.4M)
Writing Binary DB to ./ClockTreeSynthesis/_CTS.enc.dat/fifo.v.bin in single-threaded mode...
% End Save netlist data ... (date=10/28 19:49:45, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
Saving symbol-table file ...
Saving congestion map file ./ClockTreeSynthesis/_CTS.enc.dat/fifo.route.congmap.gz ...
% Begin Save AAE data ... (date=10/28 19:49:45, mem=1267.4M)
Saving AAE Data ...
% End Save AAE data ... (date=10/28 19:49:45, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
Saving preference file ./ClockTreeSynthesis/_CTS.enc.dat/gui.pref.tcl ...
Saving mode setting ...
Saving global file ...
% Begin Save floorplan data ... (date=10/28 19:49:45, mem=1267.4M)
Saving floorplan file ...
% End Save floorplan data ... (date=10/28 19:49:50, total cpu=0:00:00.0, real=0:00:06.0, peak res=1267.4M, current mem=1267.4M)
Saving PG file ./ClockTreeSynthesis/_CTS.enc.dat/fifo.pg.gz, version#2, (Created by Innovus v20.14-s095_1 on Tue Oct 28 19:49:51 2025)
*** Completed savePGFile (cpu=0:00:00.0 real=0:00:00.0 mem=1535.3M) ***
Saving Drc markers ...
... No Drc file written since there is no markers found.
% Begin Save placement data ... (date=10/28 19:49:51, mem=1267.4M)
** Saving stdCellPlacement_binary (version# 2) ...
Save Adaptive View Pruning View Names to Binary file
% End Save placement data ... (date=10/28 19:49:51, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
% Begin Save routing data ... (date=10/28 19:49:51, mem=1267.4M)
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.0 real=0:00:00.0 mem=1535.3M) ***
% End Save routing data ... (date=10/28 19:49:51, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
Saving property file ./ClockTreeSynthesis/_CTS.enc.dat/fifo.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=1538.3M) ***
#Saving pin access data to file ./ClockTreeSynthesis/_CTS.enc.dat/fifo.apa ...
#
Saving rc congestion map ./ClockTreeSynthesis/_CTS.enc.dat/fifo.congmap.gz ...
% Begin Save power constraints data ... (date=10/28 19:49:51, mem=1267.4M)
% End Save power constraints data ... (date=10/28 19:49:51, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
Generated self-contained design _CTS.enc.dat
#% End save design ... (date=10/28 19:49:52, total cpu=0:00:00.6, real=0:00:07.0, peak res=1267.4M, current mem=1267.4M)
*** Message Summary: 0 warning(s), 0 error(s)

0
innovus 2>
```



# cost cts

File Edit View Search Terminal Help

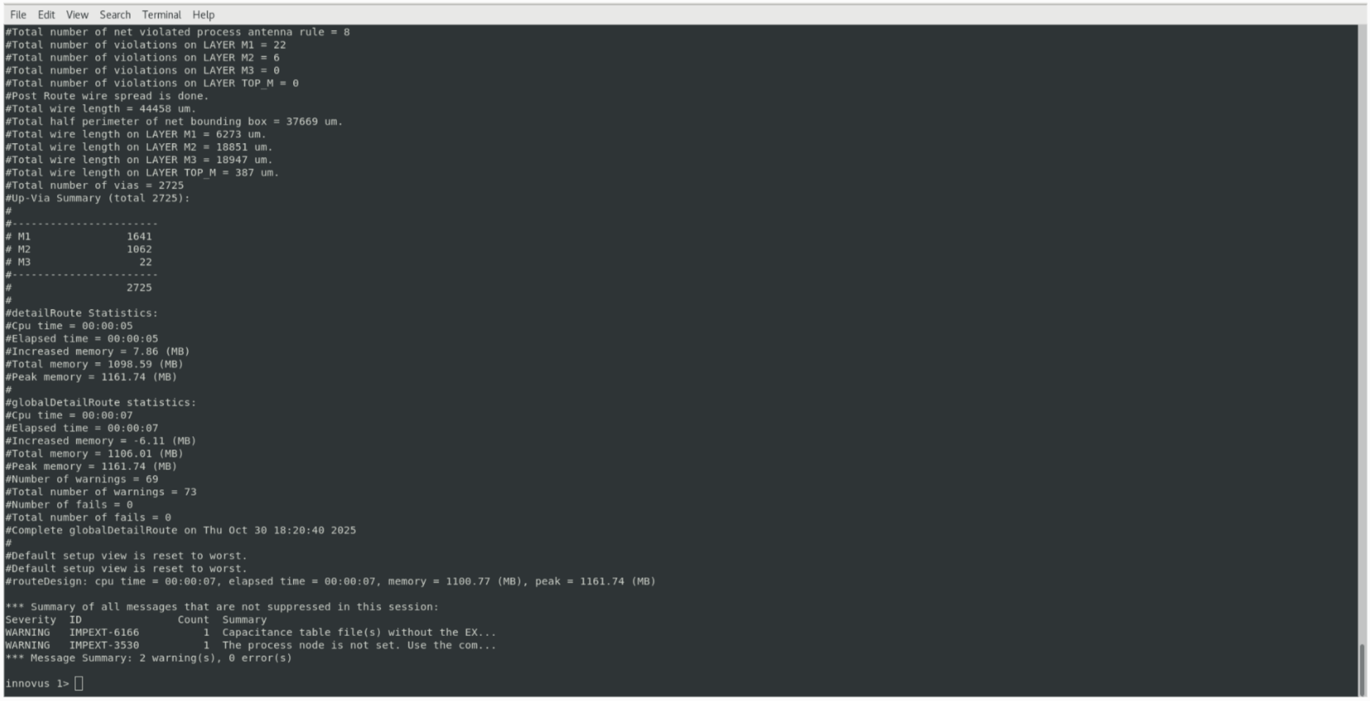
```
# Design Stage: PreRoute
# Design Name: fifo
# Design Mode: 90nm
# Analysis Mode: MMMC OCV
# Parasitics Mode: No SPEF/RCDB
# Signoff Settings: SI Off
#################################################################################
Start delay calculation (fullDC) (1 T). (MEM=1598.12)
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=1618.34 CPU=0:00:00.1 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1618.34 CPU=0:00:00.1 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.1 real=0:00:00.0 totSessionCpu=0:02:07 mem=1618.3M)

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Setup views included:
 worst

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  4.908  |  8.779  |  4.908  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   465   |   328   |   137   |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 1.318%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------------
Reported timing to dir timingReports
Total CPU time: 0.73 sec
Total Real time: 7.0 sec
Total Memory Usage: 1591.597656 Mbytes
*** timeDesign #3 [finish] : cpu/real = 0:00:00.7/0:00:07.0 (0.1), totSession cpu/real = 0:02:07.1/0:07:10.0 (0.3), mem = 1591.6M
innovus 2> 
```

# Hold voilation removed

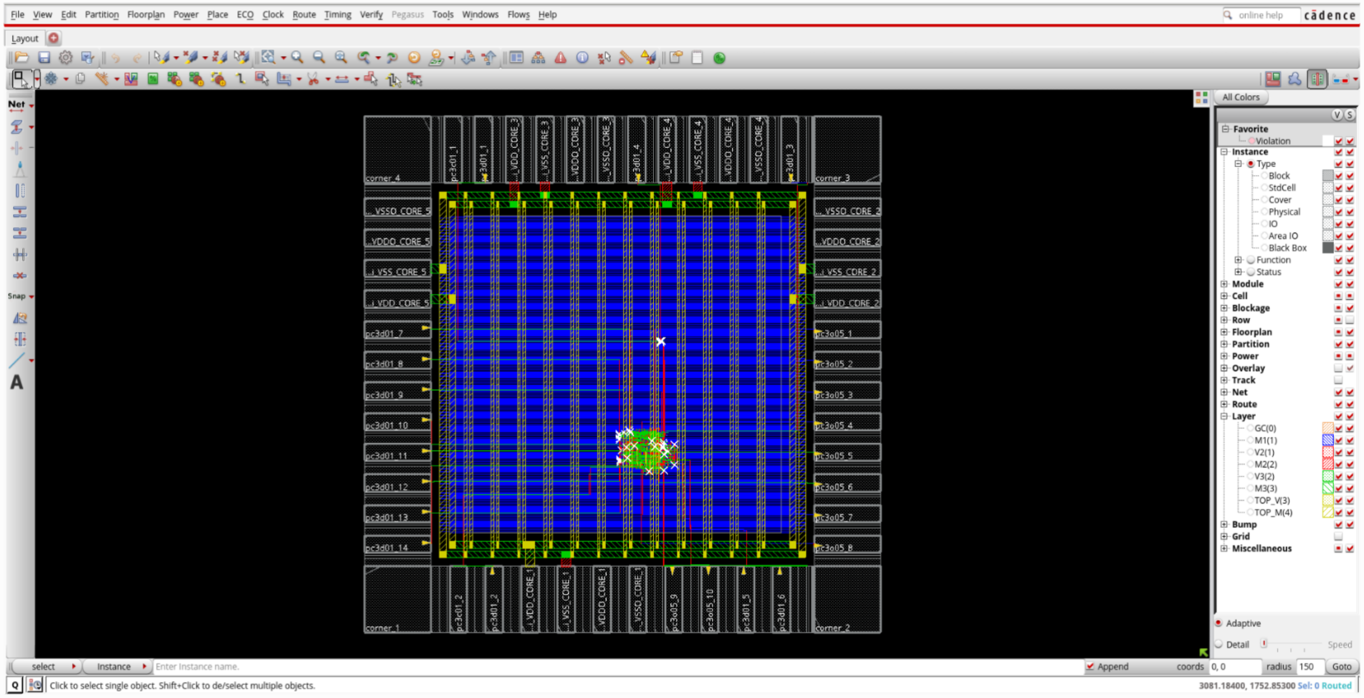
File Edit View Search Terminal Help

```
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=1704.63 CPU=0:00:00.1 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1704.63 CPU=0:00:00.1 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.2 real=0:00:00.0 totSessionCpu=0:03:03 mem=1704.6M)

------------------------------------------------------------------
     optDesign Final Summary
------------------------------------------------------------------

Setup views included:
 worst
Hold views included:
 best

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  4.908  |  8.779  |  4.908  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   465   |   328   |   137   |
+--------------------+---------+---------+---------+

+--------------------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.251  |  0.251  |  0.638  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   465   |   328   |   137   |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 1.319%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------------
**optDesign ... cpu = 0:00:05, real = 0:00:12, mem = 1414.0M, totSessionCpu=0:03:03 **
*** Finished optDesign ***
Info: Destroy the CCOpt slew target map.
*** optDesign #3 [finish] : cpu/real = 0:00:05.1/0:00:11.6 (0.4), totSession cpu/real = 0:03:03.5/0:10:41.5 (0.3), mem = 1672.9M
innovus 2> 
```

# FIFO_ Routing



```
#Total number of net violated process antenna rule = 8
#Total number of violations on LAYER M1 = 22
#Total number of violations on LAYER M2 = 6
#Total number of violations on LAYER M3 = 0
#Total number of violations on LAYER TOP_M = 0
#Post Route wire spread is done.
#Total wire length = 44458 um.
#Total half perimeter of net bounding box = 37669 um.
#Total wire length on LAYER M1 = 6273 um.
#Total wire length on LAYER M2 = 18851 um.
#Total wire length on LAYER M3 = 18947 um.
#Total wire length on LAYER TOP_M = 387 um.
#Total number of vias = 2725
#Up-Via Summary (total 2725):
#
#-----------------------
# M1               1641
# M2               1062
# M3                 22
#-----------------------
#                  2725
#
#detailRoute Statistics:
#Cpu time = 00:00:05
#Elapsed time = 00:00:05
#Increased memory = 7.86 (MB)
#Total memory = 1098.59 (MB)
#Peak memory = 1161.74 (MB)
#
#globalDetailRoute statistics:
#Cpu time = 00:00:07
#Elapsed time = 00:00:07
#Increased memory = -6.11 (MB)
#Total memory = 1106.01 (MB)
#Peak memory = 1161.74 (MB)
#Number of warnings = 69
#Total number of warnings = 73
#Number of fails = 0
#Total number of fails = 0
#Complete globalDetailRoute on Thu Oct 30 18:20:40 2025
#
#Default setup view is reset to worst.
#Default setup view is reset to worst.
#routeDesign: cpu time = 00:00:07, elapsed time = 00:00:07, memory = 1100.77 (MB), peak = 1161.74 (MB)

*** Summary of all messages that are not suppressed in this session:
Severity  ID               Count  Summary
WARNING   IMPEXT-6166          1  Capacitance table file(s) without the EX...
WARNING   IMPEXT-3530          1  The process node is not set. Use the com...
*** Message Summary: 2 warning(s), 0 error(s)

innovus 1>
```



# Post Route Timing

File Edit View Search Terminal Help

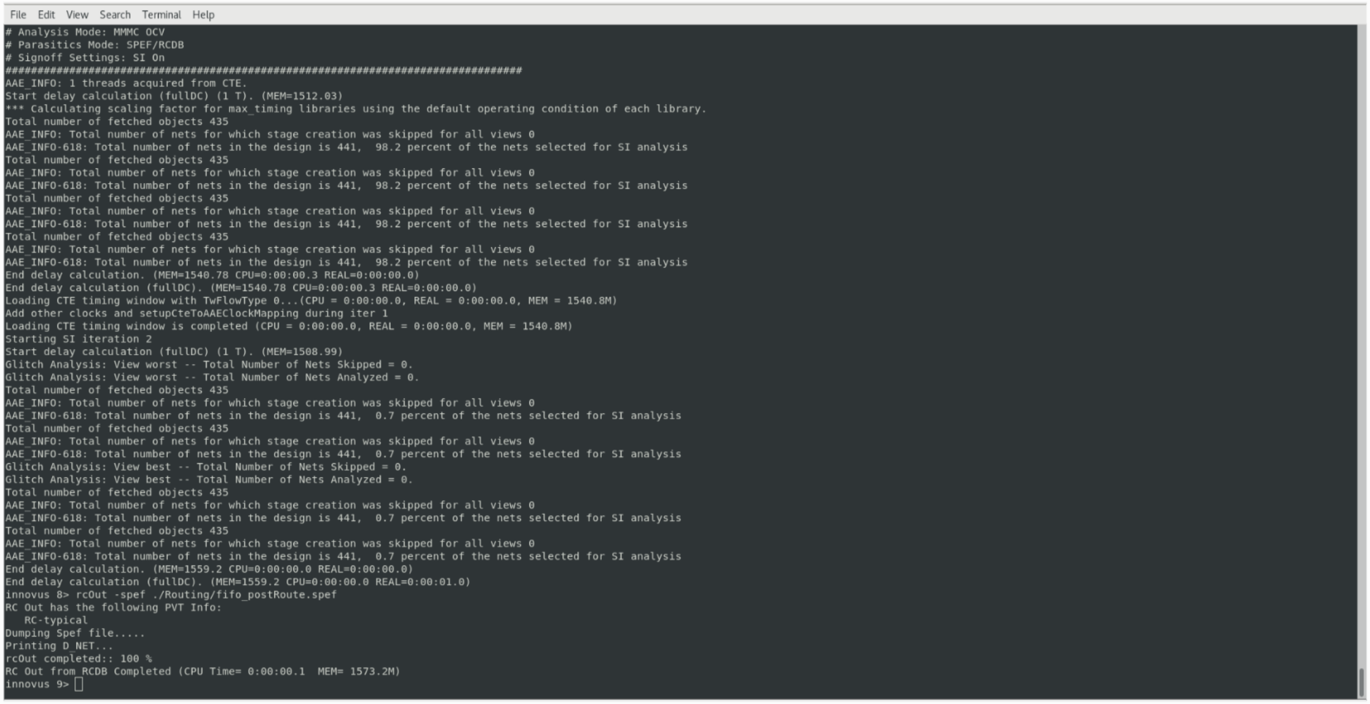
```
Loading CTE timing window is completed (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 1418.6M)
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=1384.75)
Glitch Analysis: View worst -- Total Number of Nets Skipped = 0.
Glitch Analysis: View worst -- Total Number of Nets Analyzed = 0.
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  0.7 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  0.7 percent of the nets selected for SI analysis
End delay calculation. (MEM=1433.97 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1433.97 CPU=0:00:00.0 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.6 real=0:00:01.0 totSessionCpu=0:04:18 mem=1434.0M)

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Setup views included:
 worst

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  4.890  |  8.715  |  4.890  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   465   |   328   |   137   |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 1.319%
Total number of glitch violations: 0
------------------------------------------------------------------
Reported timing to dir timingReports
Total CPU time: 1.6 sec
Total Real time: 13.0 sec
Total Memory Usage: 1423.984375 Mbytes
Reset AAE Options
*** timeDesign #1 [finish] : cpu/real = 0:00:01.6/0:00:13.5 (0.1), totSession cpu/real = 0:04:18.2/0:16:31.4 (0.3), mem = 1424.0M
innovus 1>
```

# Post route hold timing

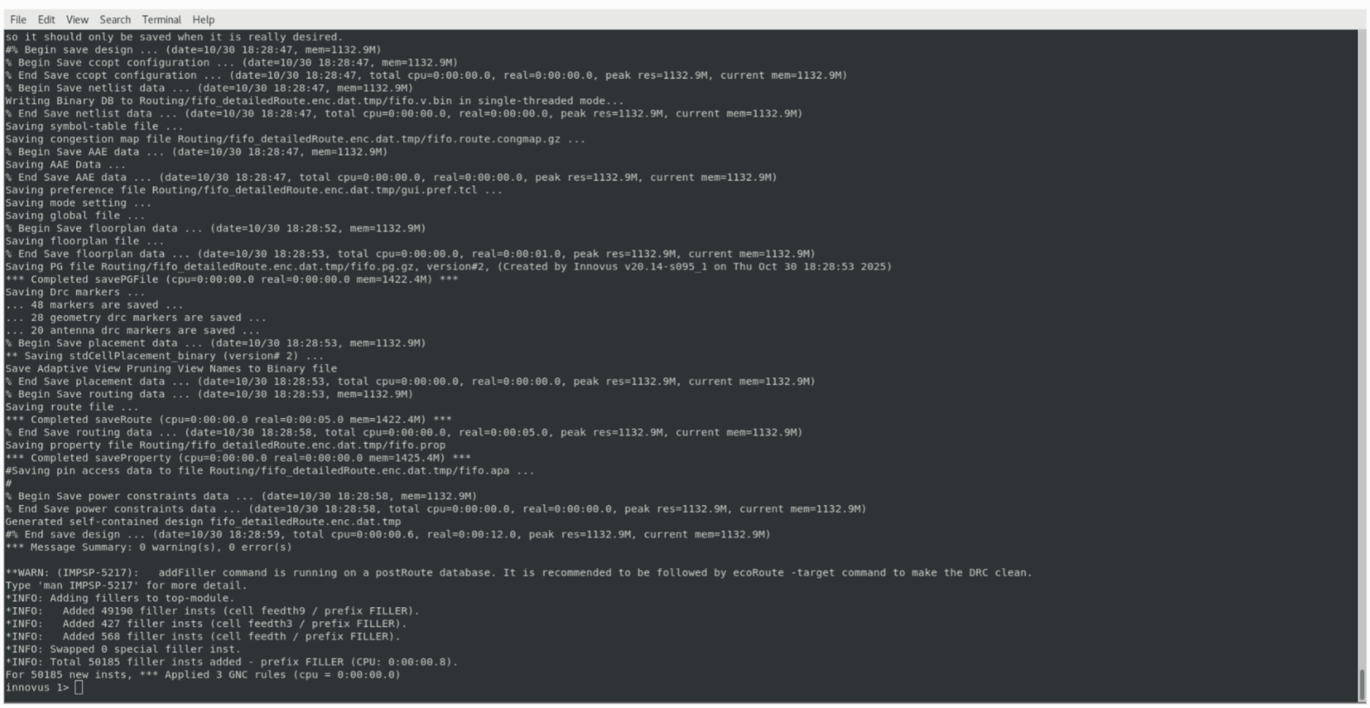
File Edit View Search Terminal Help

```
*** Calculating scaling factor for min_timing libraries using the default operating condition of each library.
Initializing multi-corner resistance tables ...
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  98.2 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  98.2 percent of the nets selected for SI analysis
End delay calculation. (MEM=1432.34 CPU=0:00:00.1 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1432.34 CPU=0:00:00.1 REAL=0:00:00.0)
Loading CTE timing window with TwFlowType 0...(CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 1432.3M)
Add other clocks and setupCteToAAEClockMapping during iter 1
Loading CTE timing window is completed (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 1432.3M)
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=1393.46)
Glitch Analysis: View best -- Total Number of Nets Skipped = 0.
Glitch Analysis: View best -- Total Number of Nets Analyzed = 0.
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  0.0 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  0.0 percent of the nets selected for SI analysis
End delay calculation. (MEM=1454.71 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1454.71 CPU=0:00:00.0 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.3 real=0:00:00.0 totSessionCpu=0:04:34 mem=1454.7M)

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Hold views included:
 best

+--------------------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.253  |  0.253  |  0.645  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   465   |   328   |   137   |
+--------------------+---------+---------+---------+

Density: 1.319%
------------------------------------------------------------------
Reported timing to dir timingReports
Total CPU time: 1.32 sec
Total Real time: 2.0 sec
Total Memory Usage: 1362.976562 Mbytes
Reset AAE Options
*** timeDesign #2 [finish] : cpu/real = 0:00:01.3/0:00:01.9 (0.7), totSession cpu/real = 0:04:34.2/0:17:34.7 (0.3), mem = 1363.0M
innovus 1>
```

# Fillers

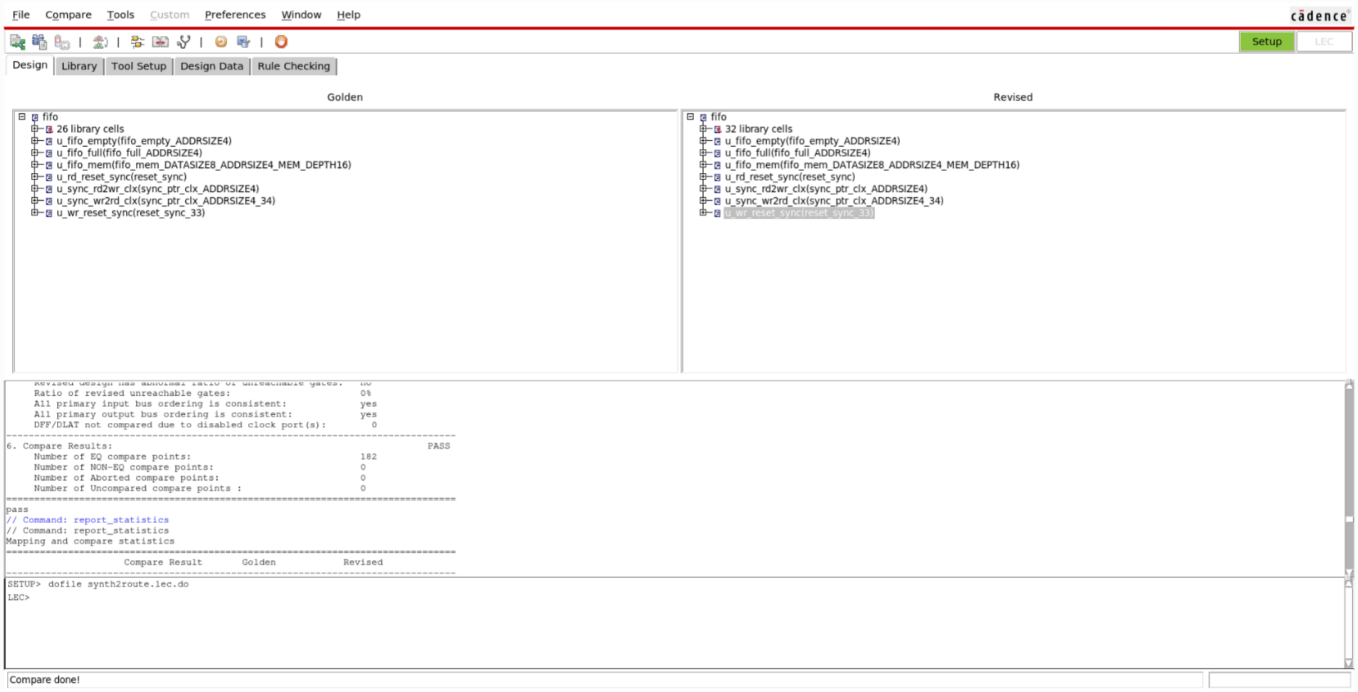



```
# Analysis Mode: MMMC OCV
# Parasitics Mode: SPEF/RCDB
# Signoff Settings: SI On
#################################################################################
AAE_INFO: 1 threads acquired from CTE.
Start delay calculation (fullDC) (1 T). (MEM=1512.03)
*** Calculating scaling factor for max_timing libraries using the default operating condition of each library.
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  98.2 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  98.2 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  98.2 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  98.2 percent of the nets selected for SI analysis
End delay calculation. (MEM=1540.78 CPU=0:00:00.3 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1540.78 CPU=0:00:00.3 REAL=0:00:00.0)
Loading CTE timing window with TwFlowType 0...(CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 1540.8M)
Add other clocks and setupCteToAAEClockMapping during iter 1
Loading CTE timing window is completed (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 1540.8M)
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=1508.99)
Glitch Analysis: View worst -- Total Number of Nets Skipped = 0.
Glitch Analysis: View worst -- Total Number of Nets Analyzed = 0.
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  0.7 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  0.7 percent of the nets selected for SI analysis
Glitch Analysis: View best -- Total Number of Nets Skipped = 0.
Glitch Analysis: View best -- Total Number of Nets Analyzed = 0.
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  0.7 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 441,  0.7 percent of the nets selected for SI analysis
End delay calculation. (MEM=1559.2 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1559.2 CPU=0:00:00.0 REAL=0:00:01.0)
innovus 8> rcOut -spef ./Routing/fifo_postRoute.spef
RC Out has the following PVT Info:
   RC-typical
Dumping Spef file.....
Printing D_NET...
rcOut completed:: 100 %
RC Out from RCDB Completed (CPU Time= 0:00:00.1  MEM= 1573.2M)
innovus 9>
```

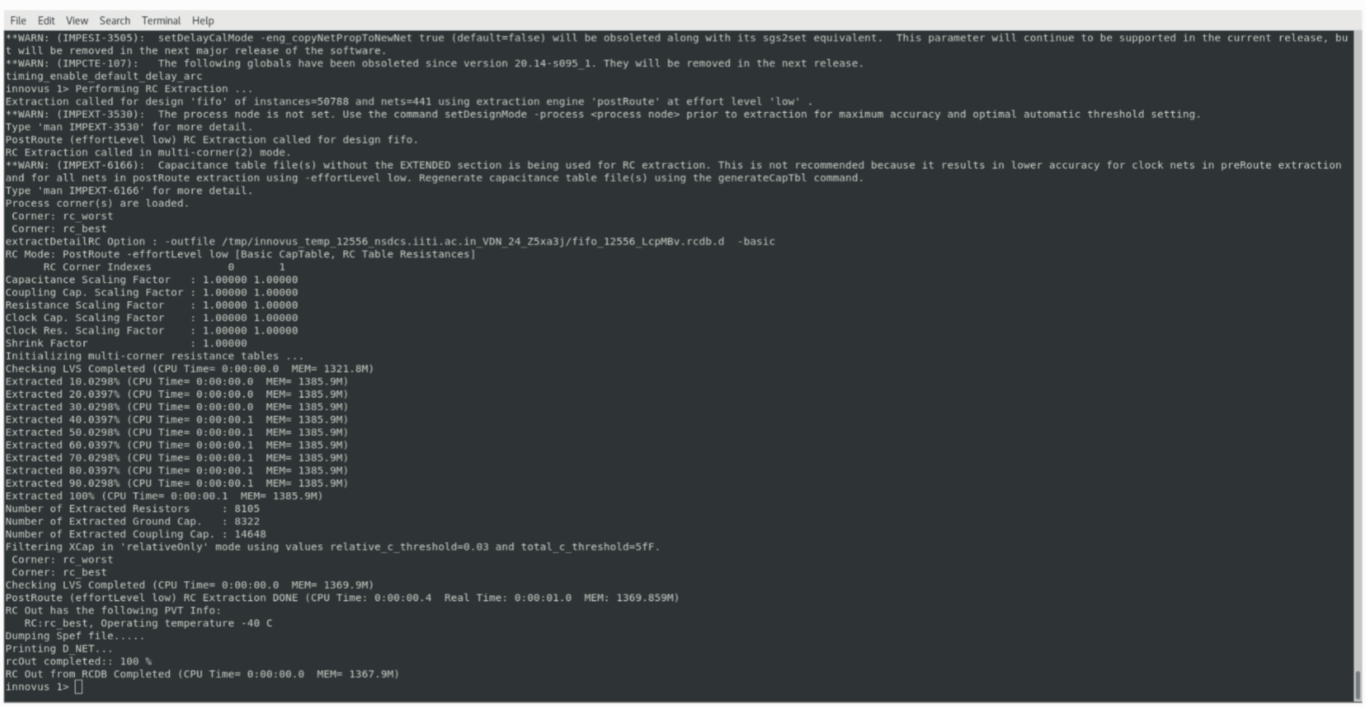



```
so it should only be saved when it is really desired.
#% Begin save design ... (date=10/30 18:28:47, mem=1132.9M)
% Begin Save ccopt configuration ... (date=10/30 18:28:47, mem=1132.9M)
% End Save ccopt configuration ... (date=10/30 18:28:47, total cpu=0:00:00.0, real=0:00:00.0, peak res=1132.9M, current mem=1132.9M)
% Begin Save netlist data ... (date=10/30 18:28:47, mem=1132.9M)
Writing Binary DB to Routing/fifo_detailedRoute.enc.dat.tmp/fifo.v.bin in single-threaded mode...
% End Save netlist data ... (date=10/30 18:28:47, total cpu=0:00:00.0, real=0:00:00.0, peak res=1132.9M, current mem=1132.9M)
Saving symbol-table file ...
Saving congestion map file Routing/fifo_detailedRoute.enc.dat.tmp/fifo.route.congmap.gz ...
% Begin Save AAE data ... (date=10/30 18:28:47, mem=1132.9M)
Saving AAE Data ...
% End Save AAE data ... (date=10/30 18:28:47, total cpu=0:00:00.0, real=0:00:00.0, peak res=1132.9M, current mem=1132.9M)
Saving preference file Routing/fifo_detailedRoute.enc.dat.tmp/gui.pref.tcl ...
Saving mode setting ...
Saving global file ...
% Begin Save floorplan data ... (date=10/30 18:28:52, mem=1132.9M)
Saving floorplan file ...
% End Save floorplan data ... (date=10/30 18:28:53, total cpu=0:00:00.0, real=0:00:01.0, peak res=1132.9M, current mem=1132.9M)
Saving PG file Routing/fifo_detailedRoute.enc.dat.tmp/fifo.pg.gz, version#2, (Created by Innovus v20.14-s095_1 on Thu Oct 30 18:28:53 2025)
*** Completed savePGFile (cpu=0:00:00.0 real=0:00:00.0 mem=1422.4M) ***
Saving Drc markers ...
... 48 markers are saved ...
... 28 geometry drc markers are saved ...
... 20 antenna drc markers are saved ...
% Begin Save placement data ... (date=10/30 18:28:53, mem=1132.9M)
** Saving stdCellPlacement_binary (version# 2) ...
Save Adaptive View Pruning View Names to Binary file
% End Save placement data ... (date=10/30 18:28:53, total cpu=0:00:00.0, real=0:00:00.0, peak res=1132.9M, current mem=1132.9M)
% Begin Save routing data ... (date=10/30 18:28:53, mem=1132.9M)
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.0 real=0:00:05.0 mem=1422.4M) ***
% End Save routing data ... (date=10/30 18:28:58, total cpu=0:00:00.0, real=0:00:05.0, peak res=1132.9M, current mem=1132.9M)
Saving property file Routing/fifo_detailedRoute.enc.dat.tmp/fifo.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=1425.4M) ***
#Saving pin access data to file Routing/fifo_detailedRoute.enc.dat.tmp/fifo.apa ...
#
% Begin Save power constraints data ... (date=10/30 18:28:58, mem=1132.9M)
% End Save power constraints data ... (date=10/30 18:28:58, total cpu=0:00:00.0, real=0:00:00.0, peak res=1132.9M, current mem=1132.9M)
Generated self-contained design fifo_detailedRoute.enc.dat.tmp
#% End save design ... (date=10/30 18:28:59, total cpu=0:00:00.6, real=0:00:12.0, peak res=1132.9M, current mem=1132.9M)
*** Message Summary: 0 warning(s), 0 error(s)

**WARN: (IMPSP-5217):   addFiller command is running on a postRoute database. It is recommended to be followed by ecoRoute -target command to make the DRC clean.
Type 'man IMPSP-5217' for more detail.
*INFO: Adding fillers to top-module.
*INFO:   Added 49190 filler insts (cell feedth9 / prefix FILLER).
*INFO:   Added 427 filler insts (cell feedth3 / prefix FILLER).
*INFO:   Added 568 filler insts (cell feedth / prefix FILLER).
*INFO: Swapped 0 special filler inst.
*INFO: Total 50185 filler insts added - prefix FILLER (CPU: 0:00:00.8).
For 50185 new insts, *** Applied 3 GNC rules (cpu = 0:00:00.0)
innovus 1>
```

# FIFO Post Route LEC




# RC Signoff



# verify Connectivity

File Edit View Search Terminal Help

```
Extracted 30.0298% (CPU Time= 0:00:00.0  MEM= 1385.9M)
Extracted 40.0397% (CPU Time= 0:00:00.1  MEM= 1385.9M)
Extracted 50.0298% (CPU Time= 0:00:00.1  MEM= 1385.9M)
Extracted 60.0397% (CPU Time= 0:00:00.1  MEM= 1385.9M)
Extracted 70.0298% (CPU Time= 0:00:00.1  MEM= 1385.9M)
Extracted 80.0397% (CPU Time= 0:00:00.1  MEM= 1385.9M)
Extracted 90.0298% (CPU Time= 0:00:00.1  MEM= 1385.9M)
Extracted 100% (CPU Time= 0:00:00.1  MEM= 1385.9M)
Number of Extracted Resistors     : 8105
Number of Extracted Ground Cap.   : 8322
Number of Extracted Coupling Cap. : 14648
Filtering XCap in 'relativeOnly' mode using values relative_c_threshold=0.03 and total_c_threshold=5fF.
 Corner: rc_worst
 Corner: rc_best
Checking LVS Completed (CPU Time= 0:00:00.0  MEM= 1369.9M)
PostRoute (effortLevel low) RC Extraction DONE (CPU Time: 0:00:00.4  Real Time: 0:00:01.0  MEM: 1369.859M)
RC Out has the following PVT Info:
   RC:rc_best, Operating temperature -40 C
Dumping Spef file.....
Printing D_NET...
rcOut completed:: 100 %
RC Out from RCDB Completed (CPU Time= 0:00:00.0  MEM= 1367.9M)
innovus 1> VERIFY_CONNECTIVITY use new engine.

******** Start: VERIFY CONNECTIVITY ********
Start Time: Thu Oct 30 19:25:48 2025

Design Name: fifo
Database Units: 1000
Design Boundary: (0.0000, 0.0000) (1939.8400, 1939.8400)
Error Limit = 1000; Warning Limit = 50
Check all nets
Net VDD_CORE: has an unconnected terminal, has special routes with opens.
Net VSS_CORE: has an unconnected terminal, has special routes with opens.
Net VDDO_CORE: no routing.
Net VSSO_CORE: no routing.

Begin Summary
    2 Problem(s) (IMPVFC-98): Net has no global routing and no special routing.
    82 Problem(s) (IMPVFC-96): Terminal(s) are not connected.
    2 Problem(s) (IMPVFC-200): Special Wires: Pieces of the net are not connected together.
    86 total info(s) created.
End Summary

End Time: Thu Oct 30 19:25:49 2025
Time Elapsed: 0:00:01.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 86 Viols.  0 Wrngs.
  (CPU Time: 0:00:00.2  MEM: 0.000M)

innovus 1> 
```

# DRC

```
File  Edit  View  Search  Terminal  Help
  VERIFY DRC ...... Sub-Area : 45 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1214.400 1214.400 1457.280 1457.280} 46 of 64
  VERIFY DRC ...... Sub-Area : 46 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1457.280 1214.400 1700.160 1457.280} 47 of 64
  VERIFY DRC ...... Sub-Area : 47 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1700.160 1214.400 1939.840 1457.280} 48 of 64
  VERIFY DRC ...... Sub-Area : 48 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {0.000 1457.280 242.880 1700.160} 49 of 64
  VERIFY DRC ...... Sub-Area : 49 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {242.880 1457.280 485.760 1700.160} 50 of 64
  VERIFY DRC ...... Sub-Area : 50 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {485.760 1457.280 728.640 1700.160} 51 of 64
  VERIFY DRC ...... Sub-Area : 51 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {728.640 1457.280 971.520 1700.160} 52 of 64
  VERIFY DRC ...... Sub-Area : 52 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {971.520 1457.280 1214.400 1700.160} 53 of 64
  VERIFY DRC ...... Sub-Area : 53 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1214.400 1457.280 1457.280 1700.160} 54 of 64
  VERIFY DRC ...... Sub-Area : 54 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1457.280 1457.280 1700.160 1700.160} 55 of 64
  VERIFY DRC ...... Sub-Area : 55 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1700.160 1457.280 1939.840 1700.160} 56 of 64
  VERIFY DRC ...... Sub-Area : 56 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {0.000 1700.160 242.880 1939.840} 57 of 64
  VERIFY DRC ...... Sub-Area : 57 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {242.880 1700.160 485.760 1939.840} 58 of 64
  VERIFY DRC ...... Sub-Area : 58 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {485.760 1700.160 728.640 1939.840} 59 of 64
  VERIFY DRC ...... Sub-Area : 59 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {728.640 1700.160 971.520 1939.840} 60 of 64
  VERIFY DRC ...... Sub-Area : 60 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {971.520 1700.160 1214.400 1939.840} 61 of 64
  VERIFY DRC ...... Sub-Area : 61 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1214.400 1700.160 1457.280 1939.840} 62 of 64
  VERIFY DRC ...... Sub-Area : 62 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1457.280 1700.160 1700.160 1939.840} 63 of 64
  VERIFY DRC ...... Sub-Area : 63 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1700.160 1700.160 1939.840 1939.840} 64 of 64
  VERIFY DRC ...... Sub-Area : 64 complete 0 Viols.

  Verification Complete : 30 Viols.

 Violation Summary By Layer and Type:

                 Short   MetSpc   Totals
       M1           17        7       24
       M2            5        1        6
       Totals       22        8       30

 *** End Verify DRC (CPU: 0:00:00.5  ELAPSED TIME: 0.00  MEM: 14.0M) ***

innovus 1> 
```

```
File  Edit  View  Search  Terminal  Help
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3d01_5' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cre
ate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3d01_6' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cre
ate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3d01_7' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cre
ate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3d01_8' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cre
ate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3d01_9' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cre
ate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3d01_10' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cr
eate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3d01_11' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cr
eate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3d01_12' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cr
eate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3d01_13' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cr
eate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3d01_14' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cr
eate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3o05_1' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cre
ate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3o05_2' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cre
ate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3o05_3' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cre
ate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520):    P/G pin 'VSSO' of instance 'pc3o05_4' is not connected to a power or ground net.  Use globalNetConnect to connect it to a power or ground net.  In a CPF-based flow, you can also use a cre
ate_global_connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (EMS-27):       Message (IMPVL-520) has exceeded the current message display limit of 20.
To increase the message display limit, refer to the product command reference manual.
Pwr name (VDD_CORE).
Pwr name (VDDO_CORE).
Gnd name (VSS_CORE).
2 Pwr names and 1 Gnd names.
innovus 3> saveNetlist ./Signoff/fifo_signoff_withPG.v
Writing Netlist "./Signoff/fifo_signoff_withPG.v" ...
innovus 4> write_sdc ./Signoff/fifo_signoff.sdc
innovus 5> 
```

# Signoff and save files

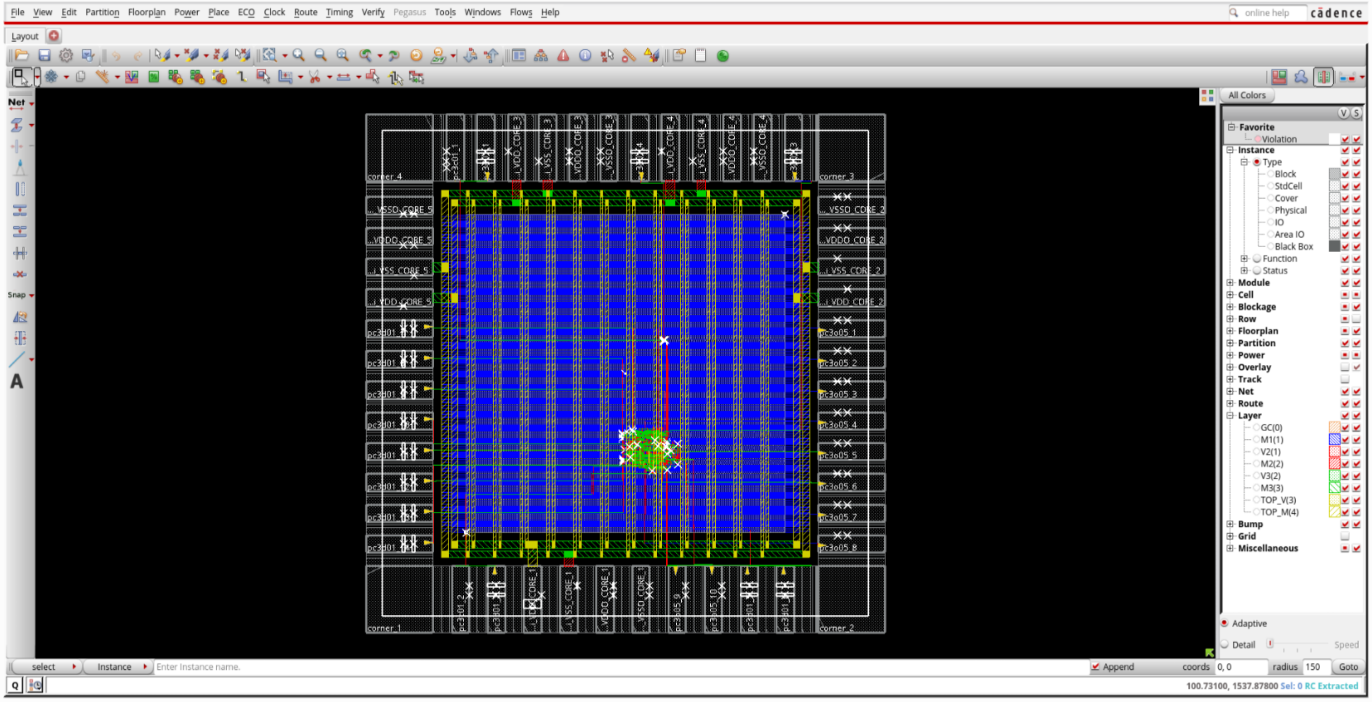
File Edit View Search Terminal Help

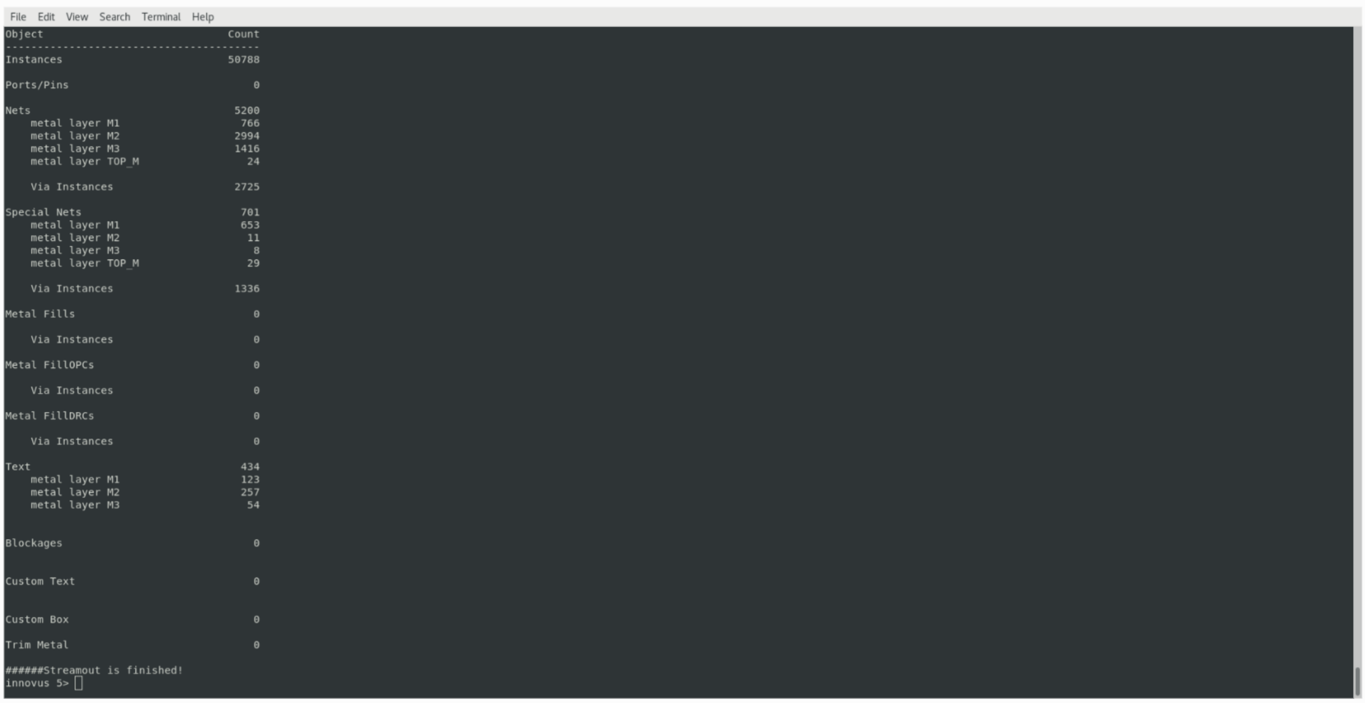
```
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=1506.6M) ***
#Saving pin access data to file Signoff/fifo_signoff.enc.dat.tmp/fifo.apa ...
% Begin Save power constraints data ... (date=10/30 19:55:37, mem=1200.2M)
% End Save power constraints data ... (date=10/30 19:55:37, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.2M, current mem=1200.2M)
Generated self-contained design fifo_signoff.enc.dat.tmp
#% End save design ... (date=10/30 19:55:38, total cpu=0:00:00.6, real=0:00:12.0, peak res=1230.9M, current mem=1200.6M)
*** Message Summary: 0 warning(s), 0 error(s)

The in-memory database contained RC information but was not saved. To save
the RC information, use saveDesign's -rc option. Note: Saving RC information can be quite large,
so it should only be saved when it is really desired.
#% Begin save design ... (date=10/30 19:55:38, mem=1200.6M)
% Begin Save ccopt configuration ... (date=10/30 19:55:38, mem=1200.6M)
% End Save ccopt configuration ... (date=10/30 19:55:38, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
% Begin Save netlist data ... (date=10/30 19:55:38, mem=1200.6M)
Writing Binary DB to Signoff/fifo_signoff.enc.dat.tmp/fifo.v.bin in single-threaded mode...
% End Save netlist data ... (date=10/30 19:55:38, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
Saving symbol-table file ...
Saving congestion map file Signoff/fifo_signoff.enc.dat.tmp/fifo.route.congmap.gz ...
% Begin Save AAE data ... (date=10/30 19:55:38, mem=1200.6M)
Saving AAE Data ...
% End Save AAE data ... (date=10/30 19:55:38, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
Saving preference file Signoff/fifo_signoff.enc.dat.tmp/gui.pref.tcl ...
Saving mode setting ...
Saving global file ...
% Begin Save floorplan data ... (date=10/30 19:55:43, mem=1200.6M)
Saving floorplan file ...
% End Save floorplan data ... (date=10/30 19:55:43, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
Saving PG file Signoff/fifo_signoff.enc.dat.tmp/fifo.pg.gz, version#2, (Created by Innovus v20.14-s095_1 on Thu Oct 30 19:55:43 2025)
*** Completed savePGFile (cpu=0:00:00.0 real=0:00:01.0 mem=1521.3M) ***
Saving Drc markers ...
... 166 markers are saved ...
... 58 geometry drc markers are saved ...
... 20 antenna drc markers are saved ...
% Begin Save placement data ... (date=10/30 19:55:44, mem=1200.6M)
** Saving stdCellPlacement_binary (version# 2) ...
Save Adaptive View Pruning View Names to Binary file
% End Save placement data ... (date=10/30 19:55:44, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
% Begin Save routing data ... (date=10/30 19:55:44, mem=1200.6M)
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.0 real=0:00:00.0 mem=1521.3M) ***
% End Save routing data ... (date=10/30 19:55:44, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
Saving property file Signoff/fifo_signoff.enc.dat.tmp/fifo.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=1524.3M) ***
#Saving pin access data to file Signoff/fifo_signoff.enc.dat.tmp/fifo.apa ...
% Begin Save power constraints data ... (date=10/30 19:55:44, mem=1200.6M)
% End Save power constraints data ... (date=10/30 19:55:44, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
Generated self-contained design fifo_signoff.enc.dat.tmp
#% End save design ... (date=10/30 19:55:44, total cpu=0:00:00.6, real=0:00:06.0, peak res=1231.4M, current mem=1200.7M)
*** Message Summary: 0 warning(s), 0 error(s)
```

File Edit View Search Terminal Help

```
Number of Input/InOut Floating Pins            : 2
Number of Output Floating Pins                 : 0
Number of Output Term Marked TieHi/Lo         *: 0

**WARN: (IMPREPO-216):  There are 20 Instances with input pins tied together.
**WARN: (IMPREPO-218):  There are 2 Floating Instance terminals.
Number of nets with tri-state drivers          : 0
Number of nets with parallel drivers           : 0
Number of nets with multiple drivers           : 0
Number of nets with no driver (No FanIn)       : 0
Number of Output Floating nets (No FanOut)     : 3
Number of High Fanout nets (>50)               : 3
**WARN: (IMPREPO-227):  There are 3 High Fanout nets (>50).
Checking routing tracks.....
Checking other grids.....
Checking FINFET Grid is on Manufacture Grid.....

Checking core/die box is on Grid.....

WARNING (IMPFP-7238): CORE's corner: (374.8800000000 , 374.8800000000) is NOT on PlacementGrid,  Please use command getFPlanMode to check current Grid settings, use command setFPlanMode to change which grid to s
nap to. And use command get_snap_grid_info to get grids' offset and pitch. Command floorplan can be used to fix this issue.
WARNING (IMPFP-7238): CORE's corner: (1564.9600000000 , 1564.9600000000) is NOT on PlacementGrid,  Please use command getFPlanMode to check current Grid settings, use command setFPlanMode to change which grid to
 snap to. And use command get_snap_grid_info to get grids' offset and pitch. Command floorplan can be used to fix this issue.
Checking snap rule ......

Checking Row is on grid......

Checking AreaIO row.....
Checking routing blockage.....
Checking components.....
Checking constraints (guide/region/fence).....
Checking groups.....
Checking Ptn Core Box.....

Checking Preroutes.....
No. of regular pre-routes not on tracks : 0
 Design check done.
Report saved in file checkDesign/fifo.main.htm.ascii

*** Summary of all messages that are not suppressed in this session:
Severity  ID               Count  Summary
WARNING   IMPDB-2136           2  %s term '%s' of %s instance '%s' does no...
WARNING   IMPREPO-227          1  There are %d High Fanout nets (>50).
WARNING   IMPREPO-205          1  There are %d Cells with missing PG PIN.
WARNING   IMPREPO-216          1  There are %d Instances with input pins t...
WARNING   IMPREPO-218          1  There are %d Floating Instance terminals...
*** Message Summary: 6 warning(s), 0 error(s)

innovus 5> The in-memory database contained RC information but was not saved. To save
the RC information, use saveDesign's -rc option. Note: Saving RC information can be quite large,
so it should only be saved when it is really desired.
#% Begin save design ... (date=10/30 19:55:14, mem=1191.7M)
% Begin Save ccopt configuration ... (date=10/30 19:55:14, mem=1193.7M)
```

# GDS Ready



File Edit View Search Terminal Help

```
Object                                 Count
----------------------------------------
Instances                              50788

Ports/Pins                                 0

Nets                                    5200
    metal layer M1                       766
    metal layer M2                      2994
    metal layer M3                      1416
    metal layer TOP_M                     24

    Via Instances                       2725

Special Nets                             701
    metal layer M1                       653
    metal layer M2                        11
    metal layer M3                         8
    metal layer TOP_M                     29

    Via Instances                       1336

Metal Fills                                0

    Via Instances                          0

Metal FillOPCs                             0

    Via Instances                          0

Metal FillDRCs                             0

    Via Instances                          0

Text                                     434
    metal layer M1                       123
    metal layer M2                       257
    metal layer M3                        54


Blockages                                  0


Custom Text                                0


Custom Box                                 0

Trim Metal                                 0

######Streamout is finished!
innovus 5>
```